



Mathematician Help Response

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Calculations/Data:

$$\begin{aligned}
 \frac{\partial^2 y}{\partial x^2} &= \frac{1}{v^2} \frac{\partial^2 y}{\partial f} \\
 \therefore y &= f(x) \frac{df}{dt} \\
 \frac{\partial y}{\partial x} &= \frac{df}{dt} \frac{\partial f}{\partial x} \\
 \frac{\partial^2 y}{\partial x^2} &= \frac{df}{dt} \frac{\partial^2 f}{\partial x^2} \\
 \text{Know } \frac{df}{dt} &= A \sin 2\pi\omega t \\
 \therefore y &= f(x) A \sin 2\pi\omega t \\
 \frac{\partial y}{\partial t} &= f(x) A 2\pi\omega \cos 2\pi\omega t \\
 \frac{\partial^2 y}{\partial t^2} &= -f(x) 4\pi^2\omega^2 A \sin 2\pi\omega t \\
 &= -4\pi^2\omega^2 f(x) \frac{df}{dt} \\
 \therefore \frac{df}{dt} \frac{\partial^2 f}{\partial x^2} &= -\frac{1}{v^2} 4\pi^2\omega^2 f(x) \frac{df}{dt} \\
 \frac{\partial^2 f}{\partial x^2} &= -\frac{4\pi^2\omega^2}{v^2} f(x) \\
 &= -\frac{4\pi^2}{\lambda^2} f(x), & \therefore v = \omega\lambda \\
 \frac{\partial^2 \psi}{\partial x^2} &= -\frac{4\pi^2}{\lambda^2} \psi \\
 &= -\frac{4\pi^2 m^2 v^2}{h^2} \psi, & \therefore \lambda = \frac{h}{mv}
 \end{aligned}$$

$$\begin{aligned} &= \frac{8\pi^2 m}{h^2} (E - V) \psi, \\ \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} &= \frac{8\pi^2 m}{h^2} (E - V) \psi, \end{aligned} \quad \begin{aligned} E &= \frac{mv^2}{2} + V \\ \text{For } \Psi(x, y, z) \end{aligned}$$

Description:

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

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